

Secure Software Engineering – 20CYS401

Lab Exercise - 4

Requirements analysis and Analysis modelling using UML Diagrams

Learning Objectives:

- Translating a natural-language requirements scenario into actors, use cases, and their relationships.
- Identifying the structural elements of a system (classes, attributes, operations, relationships) and its workflow (steps, decisions, responsibilities).
- Modelling the order of messages exchanged between objects for a specific use case scenario using a sequence diagram.

The Scenario: PayCloud HR System

PayCloud is a cloud-based payroll system used by companies. Employees log in to view and download their payslips. HR administrators upload payroll data every month for their employees. The system checks the data for errors before it is approved. Once approved, payslips are generated, and payments are sent through a payment gateway.

Tasks:

1. Draw a Use Case Diagram for the given scenario.
2. Draw a Class Diagram for the given scenario.
3. Draw an Activity Diagram for the given scenario.
4. Draw a Sequence Diagram for the given scenario.

Tool : draw.io - <https://app.diagrams.net>

References:

1. <https://se-iitkgp.vlabs.ac.in/exp/modeling-uml/>

2. <https://se-iitkgp.vlabs.ac.in/exp/uml-class-sequence/>
3. <https://se-iitkgp.vlabs.ac.in/exp/statechart-activity-modeling/>
4. <https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-uml/>